> **Human-reviewed version.** Based on the original output of run 5 of the Tech Docs Research workflow (26 September 2026). The corrections were made after the run, outside the workflow, by checking every claim against the two cited MDN pages on 16 October 2026, and were approved by Anas Mansour. The original output is kept unchanged as private evidence.

# HTTP caching with Cache-Control: Technical Brief

**Question:** Which Cache-Control directives should an API use to control caching, and how do no-cache, no-store and max-age differ?
**Audience:** Tech lead

## TL;DR

`max-age` sets how long a response stays fresh, counted in seconds from when the origin server generated it. `no-cache` lets caches store a response but requires revalidation with the origin server before each reuse; `no-store` tells caches not to store it at all. `no-cache` does not mean "don't cache". [S1]

## Key facts

- `max-age=N`: the response stays fresh until N seconds after it was generated on the origin server, not after it was received. [S1]
- `no-cache`: the response can be stored, but must be validated with the origin server before each reuse, even when the cache is disconnected from the origin server. [S1]
- `no-store`: tells caches of any kind (private or shared) not to store the response. [S1]
- If the "don't cache" you want is really "don't store", `no-store` is the directive to use. [S1]
- `s-maxage`: sets how long the response stays fresh in a shared cache (for example a proxy or CDN). Private caches ignore it; in shared caches it overrides `max-age`. [S1]
- `must-revalidate`: the response can be reused while fresh; once stale, it must be validated with the origin server before reuse. It is typically used with `max-age`. [S1]
- `private`: the response can be stored only in a private cache, such as the browser's. MDN recommends it for user-personalized content, especially responses received after login. [S1]

## How it works / how to use it

- Multiple directives are comma-separated; directives are case-insensitive, but lowercase is recommended. [S1]
- To have caches always check for updates while still reusing stored content, use `no-cache`. [S1]
- If a response should not be stored in any cache, use `no-store`. [S1][S2]
- For personalized or logged-in content, add `private`, so the response is not stored in a shared cache and reused for other users. [S1][S2]
- `s-maxage` applies to shared caches only and, there, takes precedence over `max-age`. [S1]

## Caveats & conflicts

- `max-age` counts from generation on the origin server, not from when a cache received the response. [S1]
- If both `Expires` and `Cache-Control: max-age` are present, `max-age` is preferred. [S2]
- `must-revalidate` requires revalidation only once a response is stale; a fresh response can be reused. [S1]

## Open questions

Not answered by the two sources (unchanged from the original output):

- Specific directive combinations for read-only vs. mutable API endpoints
- Optimal caching strategies for authenticated vs. public data scenarios
- Performance implications of each directive choice for API workloads
- Behavioural differences for no-cache with browser history navigation

## Sources

1. S1 - https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Cache-Control (retrieved by the workflow on 26 September 2026; re-checked 16 October 2026)
2. S2 - https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Caching (same dates)

## What changed from the original output

| Original output | Reviewed version | Why |
|---|---|---|
| no-cache must be revalidated "before each reuse, even offline" | "…even when the cache is disconnected from the origin server" | S1's wording. "Offline" could be read as the user's device being offline |
| no-store "prevents any cache from storing the response" | "tells caches of any kind … not to store the response" | S1 says caches "should not store this response": an instruction to caches |
| "For sensitive data that must not persist, use no-store" | "If a response should not be stored in any cache, use no-store" | Neither source mentions sensitive data; both say to use no-store if you don't want a response stored |
| s-maxage: "combine both when client and intermediate caching differ" | Removed; only the documented mechanism kept | Neither source recommends combining s-maxage with max-age (flagged in the 26 September review) |
| s-maxage "overrides max-age for proxies and CDNs" | Kept, worded as shared caches " (for example a proxy or CDN)" | S1 gives "Proxy, CDN" as examples of shared caches |
| Footer with the internal run ID | Removed | Internal identifier, not needed publicly |

*Original output generated by the Tech Docs Research workflow v2 on 26 September 2026. This version was corrected outside the workflow against the cited sources and approved by Anas Mansour.*